# Wycliffe Kipleting Rotich

**Platform Engineer | Distributed Systems | Python, AWS, Kubernetes**

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github.com/wycliffRotich-dev/aethergrid | Live demo: aethergrid-dashboard.onrender.com

Open to remote roles
Available to start immediately

| 7+ yrs | 876 | 75 | 29 s to 0.124 s |
|---|---|---|---|
| backend and distributed systems in production | passing tests in AetherGrid | architecture decision records (ADRs) | worst-case latency during a database outage |

## PROFESSIONAL SUMMARY

Platform engineer with 7+ years building backend and distributed systems on cloud platforms, focused on concurrency, race-condition debugging, execution ownership and failure recovery. Creator and maintainer of **AetherGrid**, an open-source distributed AI workload orchestrator covering scheduling, lease-based execution ownership, reconciliation, scoped API keys, hardened containers and tenant-scoped data access. Previously at Toyota and the Kenya Revenue Authority, working on systems that handle vehicle telemetry and national-scale financial data.

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages** | Python, TypeScript, JavaScript, SQL |
| **Backend** | FastAPI, REST APIs, Async Python, Pydantic, PostgreSQL, psycopg, SQLite, Node.js |
| **Distributed systems** | Scheduling, leases, reconciliation, concurrency, race-condition debugging, retries with backoff and jitter, circuit breakers, idempotency, worker heartbeats, node draining, Kafka |
| **Cloud and DevOps** | AWS, ECS, Kubernetes, Terraform, Docker, Docker Compose, Docker Secrets, GitHub Actions, CI/CD, Dependabot |
| **Security** | API key authentication, scoped authorization, tenant isolation, container hardening (non-root, read-only filesystem, dropped capabilities) |
| **Frontend** | React 19, TypeScript, Vite, TailwindCSS, client-side routing, Playwright |
| **Testing** | TDD, pytest, contract, integration, API and end-to-end testing, mutation verification |
| **Design** | Clean Architecture, DDD, SOLID, aggregates, value objects, state machines, repository pattern, domain events, ADRs |
| **AI and ML** | LLM evaluation, code quality assessment, AI training pipelines, data annotation and QA |

## PROFESSIONAL EXPERIENCE

### Founder and Platform Engineer

Jan 2026 - Present

AetherGrid | Open-source AI workload orchestrator | Nairobi

- Designed and built AetherGrid end to end with Python, FastAPI, PostgreSQL (raw psycopg, no ORM), React, TypeScript, Docker Compose and GitHub Actions. Four layers with a domain core that has no infrastructure imports. **876 passing tests** across unit, contract, integration, API and end-to-end, and **75 ADRs**.
- Built a multi-dimensional best-fit scheduler across CPU, memory and VRAM, with time-bound leases renewed for the full run of a job. Reconciliation reclaims work from dead workers, expired leases, offline nodes and jobs abandoned mid-cancellation, within a bounded retry budget.
- Found and closed a recurring **race-condition class** by fencing lease release, outcome reporting and renewal on lease identity, then fencing reconciliation's own deletes by expiry. Each fix began with a test that reproduced the bug and was verified against real PostgreSQL.
- Built a real subprocess engine with enforced timeouts and two-stage termination (SIGTERM, then SIGKILL), with cancellation delivered over the lease-renewal channel and an explicit CANCELLING state. Built a standalone worker agent with capped exponential backoff and full jitter that re-registers after control plane restarts, one API key per agent.
- Kept the API responsive during database outages by moving reconciliation off the event loop, splitting liveness from readiness and bounding the pool timeout to 5 seconds. Worst-case latency on an unrelated endpoint dropped from **about 29 s to 0.124 s**, measured against a stopped Postgres container.
- Secured the API with key authentication on every route, scoped keys (jobs:execute, keys:manage), key ownership and locally minted first keys. Hardened containers: non-root, read-only root filesystem, all capabilities dropped, no-new-privileges, pinned digests, database password as a Docker secret. A CI workflow asserts each property, and each check was proven to fail on a deliberate break.
- Implemented **tenant scoping** for API keys, nodes and jobs. The tenant comes from the credential, never the request, and another tenant's resource answers like a missing one. Workers, leases and events are the next stage.

- Built a resilient HTTP client (retries, circuit breaker, OAuth2, mTLS, replay of only safe requests) and a React 19 operator dashboard with a live event feed, per-job history and a Playwright end-to-end test in CI.

### LLM Technical Trainer / AI Training Specialist
Aug 2025 - Present

Mercor | Remote

- Evaluate LLM-generated code for correctness, architecture, concurrency and API behavior.
- Review backend implementations where the happy path works but the system breaks under real conditions, finding issues in state management, async execution and race conditions.
- Write technical feedback that improves model reasoning and code quality.

### AI Data Specialist, Video Annotation and Quality Review
Dec 2024 - May 2025

Atlas Capture

- Annotated and reviewed video datasets for machine-learning workflows using CVAT, Labelbox, Scale, V7 and Encord.
- Performed QA reviews and resolved inconsistent or ambiguous annotations.

### Software Engineer, Cloud Platforms
Jan 2024 - Nov 2024

Toyota | Nairobi

- Fixed concurrency problems in worker queues, resolving race conditions in Kafka-based ingestion pipelines for vehicle and telematics event data.
- Built high-volume Kafka data ingestion pipelines with asynchronous downstream processing.
- Deployed and orchestrated containerized workloads on Kubernetes and ECS across AWS environments.
- Provisioned cloud infrastructure with Terraform, improving resource allocation and reducing provisioning time.
- Designed REST APIs with strict DTO validation and refactored tightly coupled components to separate business logic from framework code.

### Backend Cloud Engineer
Nov 2018 - Oct 2023

Kenya Revenue Authority (KRA) | Nairobi

- Worked with senior engineers on production revenue systems handling real financial data at national scale.
- Traced race conditions where concurrent requests interfered with shared state, isolating root causes across application logic, shared state and database interactions.
- Investigated timing-dependent bugs that were difficult to reproduce through standard request-response testing.

## EDUCATION

### Master of Science, Computer Science
Jan 2022 - Dec 2024

United States International University-Africa (USIU-Africa), part-time

### Bachelor of Engineering, Computer Software Engineering
Aug 2014 - Nov 2018

The Technical University of Kenya